separation principles via optimal transport

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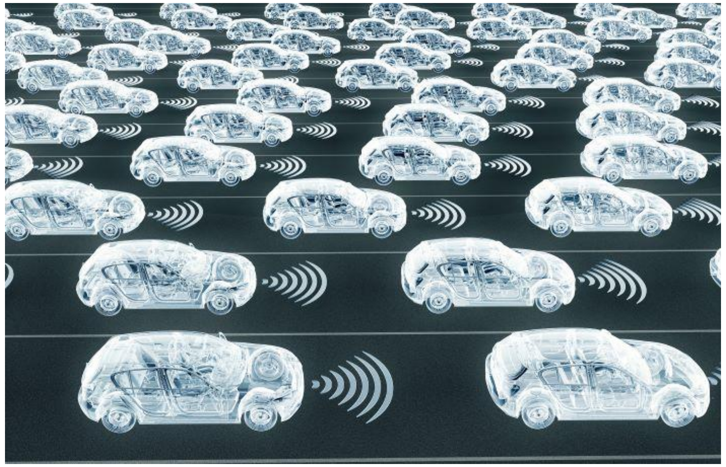
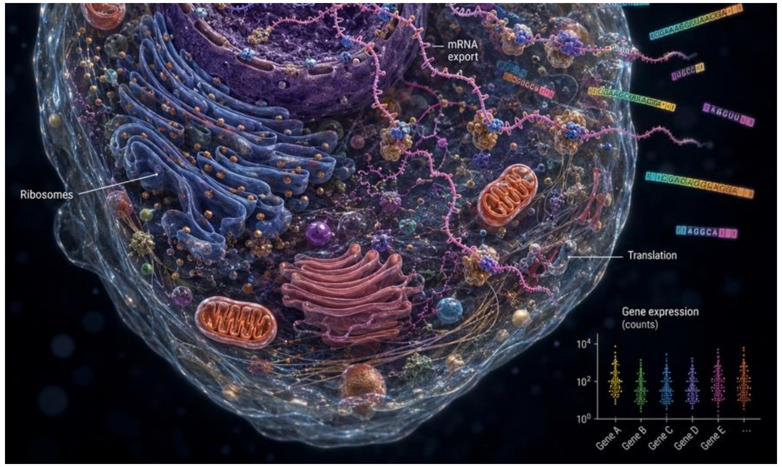
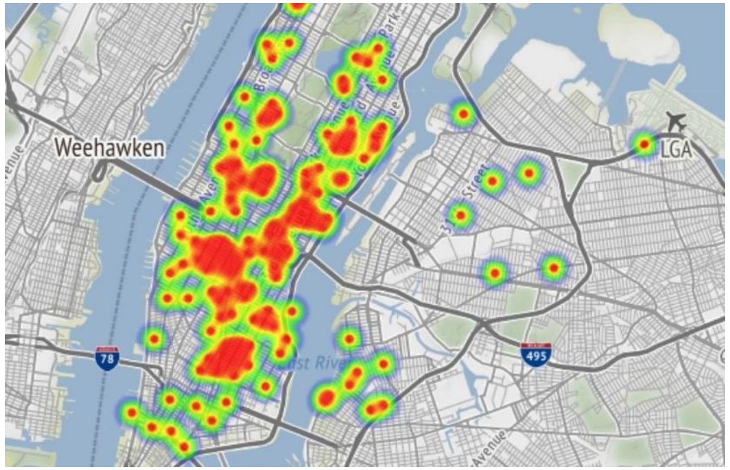
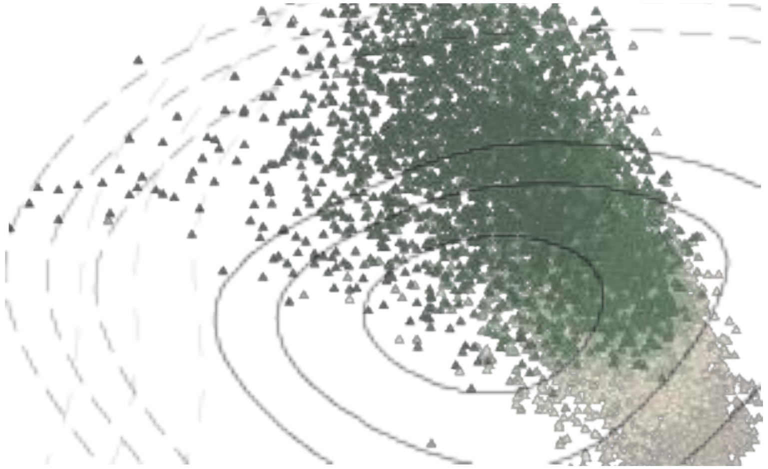
*Sorry! Nicolas will present on Friday the variational structure of the Wasserstein space

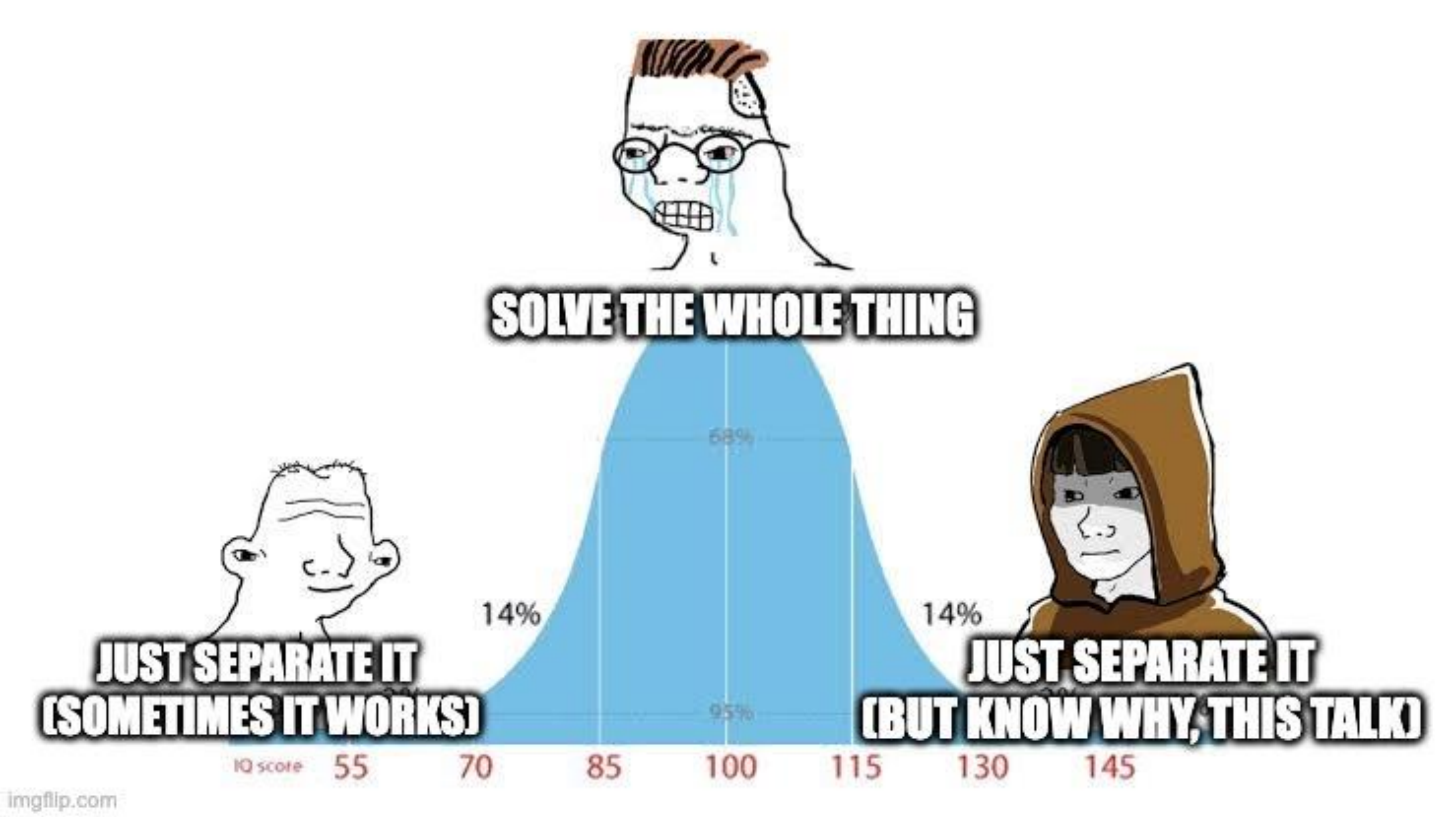
real world complexity demands shortcuts



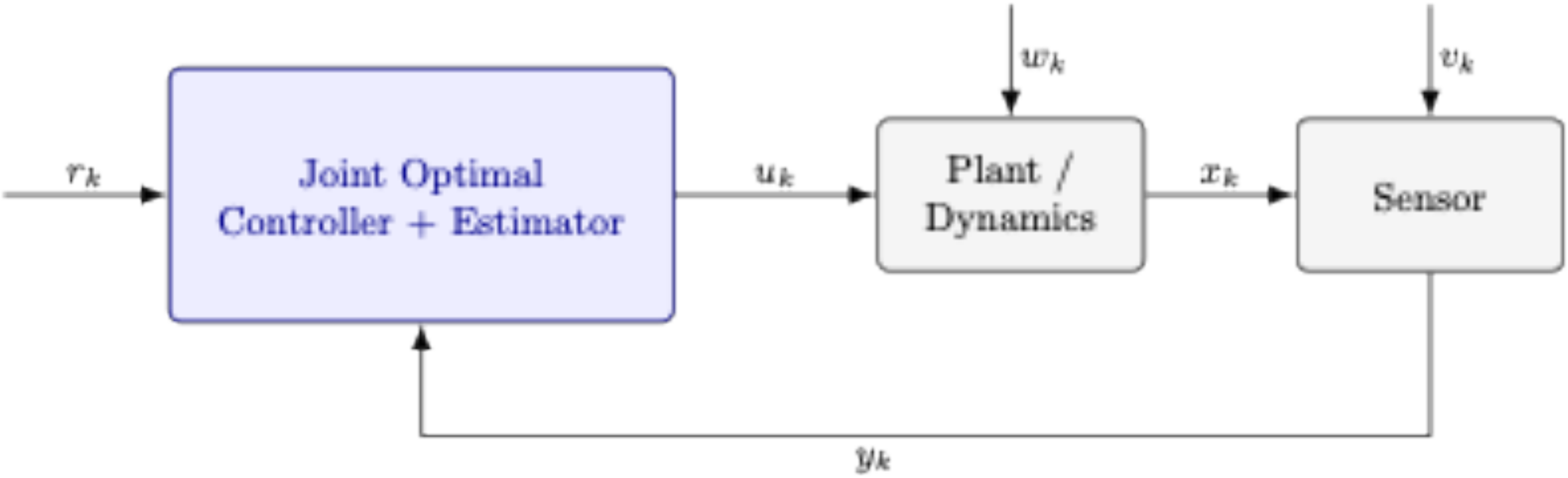




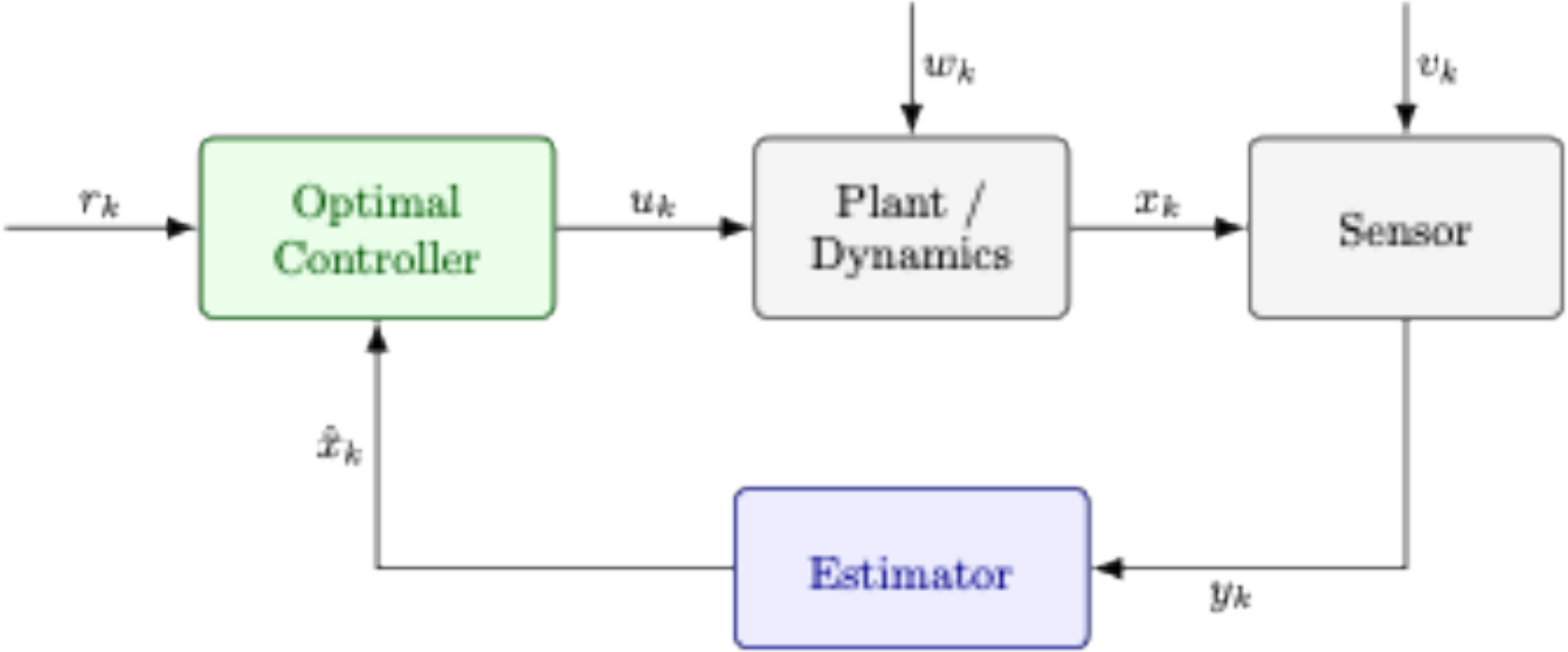



SOLVE THE WHOLE THING
68%
14%
14%
JUST SEPARATE IT
(SOMETIMES IT WORKS)
JUST SEPARATE IT
(BUT KNOW WHY, THIS TALK)
95%
IQ score 55 70 85 100 115 130 145
imgflip.com

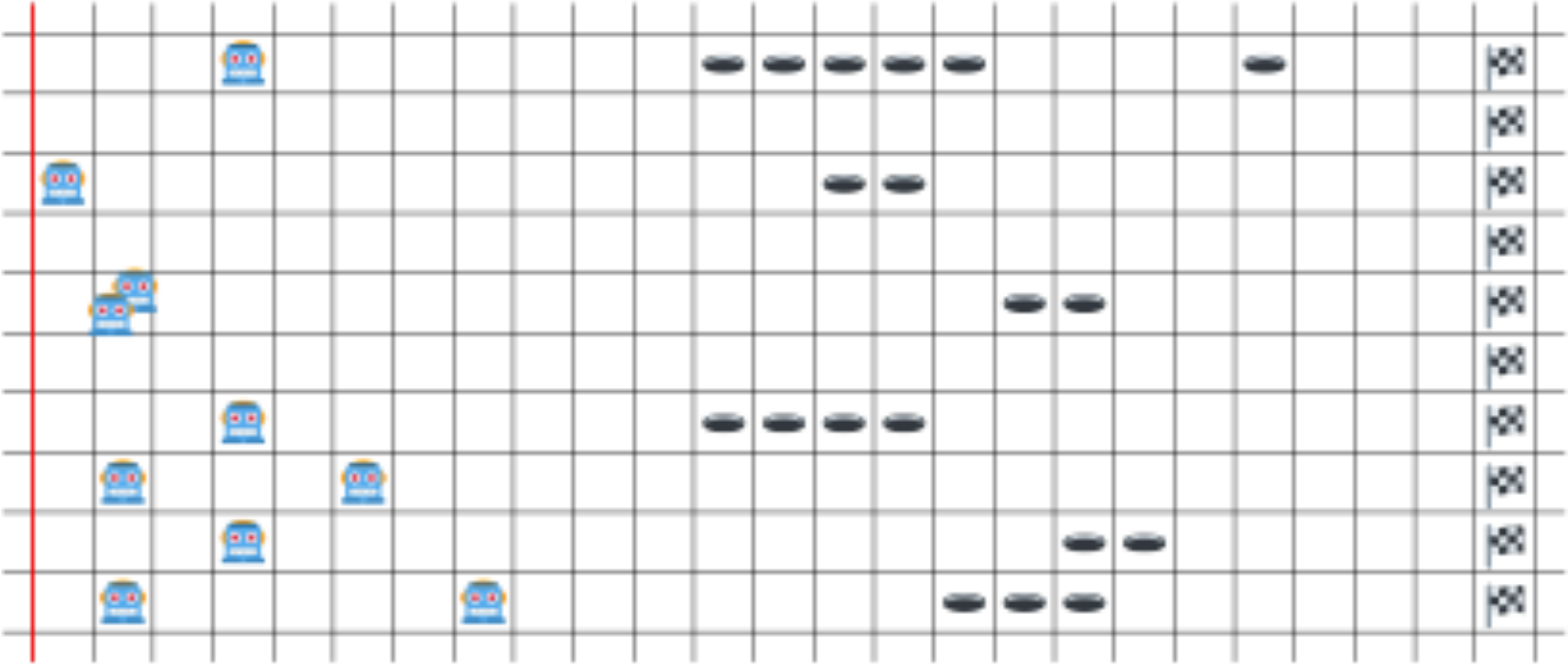
prominent example: LQG

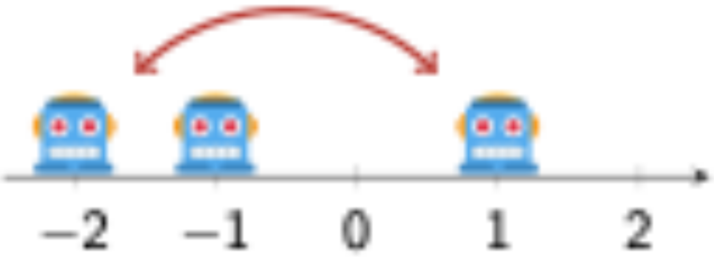


prominent example: LQG



a separation principle for multi-agent control



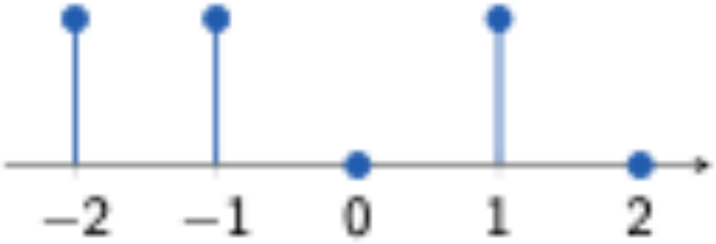


$$x = \begin{bmatrix} \text{state agent 1} \\ \text{state agent 2} \\ \text{state agent 3} \end{bmatrix} \neq \begin{bmatrix} \text{state agent 1} \\ \text{state agent 3} \\ \text{state agent 2} \end{bmatrix}$$

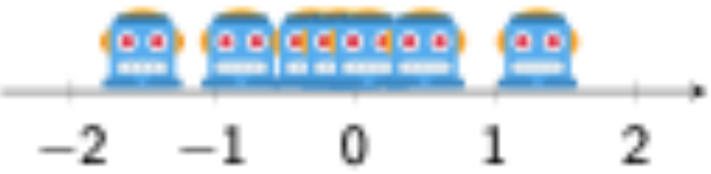
… and this happens because we artificially label the agents.

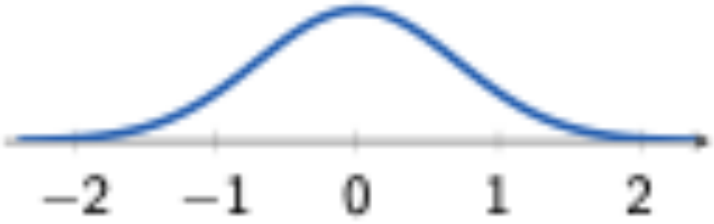
The remedy: Model a fleet via a distribution!

Finite fleet



Infinite fleet





Terminal cost (formation cost):
optimal transport discrepancy with transportation cost $g_N : X_N \times R \to \mathbb{R}_{\geq 0}$ from a reference ρ

Stage cost (aggregate cost):
average cost over the fleet
$g : X_k \times U_k \to \mathbb{R}_{\geq 0}$

$$
\begin{aligned}
J(\mu, \rho) = \inf_{\substack{\mu_k \in \mathcal{P}(X_k) \\ \lambda_k \in \mathcal{P}(X_k \times U_k)}} \quad & \mathcal{K}[g_N](\mu_N, \rho) + \sum_{k=0}^{N-1} \mathbb{E}^{\lambda_k}[g_k] \\
\text{s.t.} \quad & \mu_{k+1} = f_{k\#}\lambda_k, \\
& \mu_0 = \mu, \\
& \left(\mathrm{proj}_{X_k}^{X_k \times U_k}\right)_{\#} \lambda_k = \mu_k.
\end{aligned}
$$

State distribution μ_k:
distribution of agents

State-input distribution λ_k:
probability of being at a given state and use a given action

Dynamics:
$f_k : X_k \times U_k \to X_{k+1}$
dynamics of each agent

Challenge:

We can use dynamic programming,

but the cost-to-go and feedback laws are arbitrary functions over distributions.

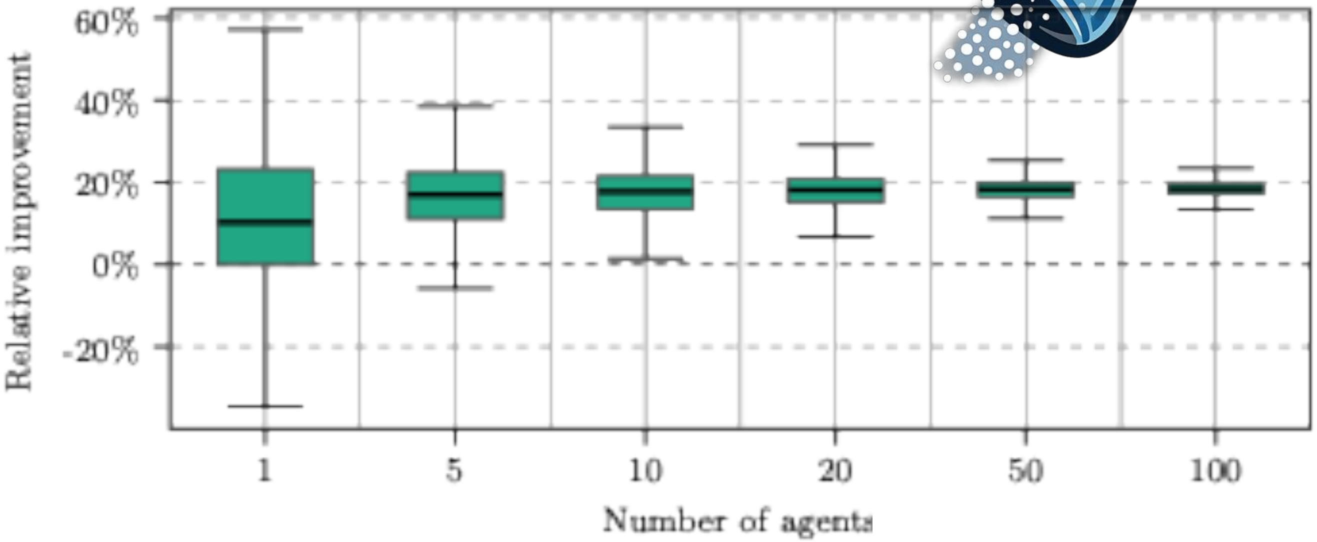
Theorem: A separation principle holds true:

1. assign agents to targets solving a fleet-level dispatching problem;
2. route individual agents.

For the optimal transport aficionados: At every stage k:

1. The cost-to-go is an optimal transport problem:
$$J_k(\mu_k, \rho) = \mathcal{K}[j_k](\mu_k, \rho) = \inf_{\gamma \in \Gamma(\mu_k, \rho)} \int_{X_k \times R} j_k(x_k, r) \mathrm{d}\gamma(x_k, r),$$
where j_k is a single-agent cost-to-go:
$$j_N(x_N, r) := g_N(x_N, r); \qquad j_k(x_k, r) := \inf_{u_k \in U_k} g_k(x_k, u_k) + j_{k+1}(f_k(x_k, u_k), r).$$
2. Optimal state-input distribution:
$$\lambda_k^* = \left(\mathrm{proj}_{X_k}^{X_k \times R}, u_k^*\right)_{\#} \gamma_k^*$$
where u_k^* is the optimal single-agent control input and γ_k^* is the optimal dispatching plan in the cost-to-go $J_k(\mu_k, \rho)$.

… with consequences for learning



Algorithm 1 Separation-based Assignment and Learning via optimal Transport

for each episode **do**
  Sample starting states and targets.
  Match agents to targets via optimal transport.
  Act: roll out the shared policy with assignments fixed.
  Learn: update the shared policy and value from pooled trajectories.
end for

a separation principle for multi-agent learning

diffusion processes in Switzerland

goal: identify the population dynamics from population data

find the energy functional that best explains the data

Wasserstein distance

$$\mu_{t+1} = \underset{\mu \in \mathcal{P}(\mathbb{R}^d)}{\operatorname{argmin}} J(\mu) + \frac{1}{2\tau} W_2(\mu, \mu_t)^2$$

discretization

population data at time t+1
(given)

population distribution at time t
(given)

$$J(\mu) = \int_{\mathbb{R}^d} V(x) \mathrm{d}\mu(x) + \beta \int_{\mathbb{R}^d} \log(\rho(x))\rho(x) \mathrm{d}x \quad \Leftrightarrow \quad \mathrm{d}X_t = -\nabla V(X)\mathrm{d}t + \sqrt{2\beta}\,\mathrm{d}W_t$$

$$\mu_{t+1} = \underset{\mu \in \mathcal{P}(\mathbb{R}^d)}{\operatorname{argmin}} J(\mu) + \frac{1}{2\tau} W_2(\mu, \mu_t)^2$$

first-order optimality conditions in the Wasserstein space

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \left\| \nabla J(\mu_{t+1})(x_{t+1}) + \frac{1}{\tau}(x_{t+1} - x_t) \right\|^2 d\gamma_t(x_t, x_{t+1}) = 0$$

optimal coupling between and (only data-dependent)

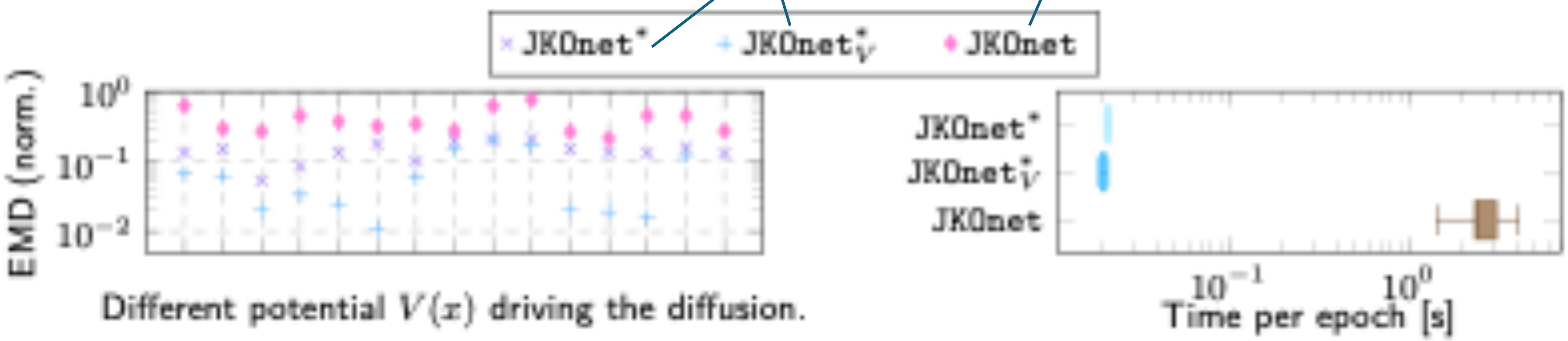
loss function for estimation

$$\min_{\theta} \sum_{t=0}^{T-1} \int_{\mathbb{R}^d \times \mathbb{R}^d} \left\| \nabla J_\theta(\mu_{t+1})(x_{t+1}) + \frac{1}{\tau}(x_{t+1} - x_t) \right\|^2 d\gamma_t(x_t, x_{t+1}) = 0$$

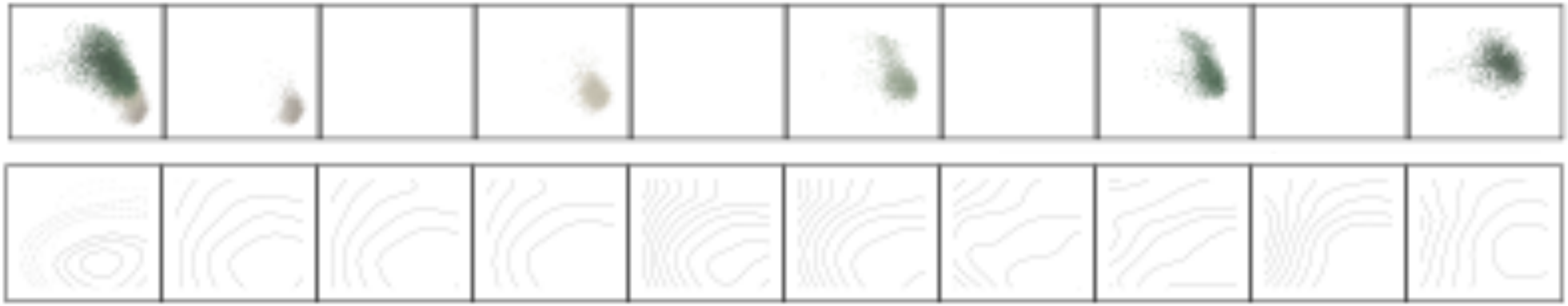
Algorithm 1 JKOnet*: Learning diffusion at lightspeed

```
for each training step do
    Sample consecutive population snapshots.
    Match particles across snapshots via optimal transport.
    Learn: update the energy by minimizing the MSE between predicted and observed displacements.
end for
```

results



RNA-seq



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